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Relationship between social media addiction, sleep quality, and occupational balance among Generation Z university students

Hülya Yaman Dağdelen^{1*} and Zeynep Bahadır²

Abstract

Objectives The widespread integration of social media into daily life has raised concerns about its impact on young adults' well-being. This cross-sectional study examined the relationship between social media addiction, sleep quality, and occupational balance among Generation Z university students in Türkiye.

Methods The sample included 136 undergraduates born in or after 2000. Data were collected using the Social Media Addiction Scale, the Pittsburgh Sleep Quality Index, and the Occupational Balance Questionnaire.

Results Higher social media addiction scores were weakly associated with poorer sleep quality ($r = .219$, FDR-adjusted $p = .016$) and higher occupational balance scores ($r = .275$, FDR-adjusted $p = .003$). Social media addiction scores differed significantly according to daily usage duration, with students reporting more than six hours of daily social media use having the highest addiction scores ($F = 7.57$, $p < .001$). In the regression analysis, social media addiction remained significantly associated with poorer sleep quality after controlling for age, gender, academic year, and daily social media use duration ($\beta = 0.255$, $p = .007$). However, social media addiction was not a significant predictor of occupational balance after adjustment, whereas sleep quality was significantly associated with occupational balance ($\beta = 0.370$, $p < .001$).

Conclusion Social media addiction appears to be associated with sleep quality among Generation Z university students, while its relationship with occupational balance seems more complex and should be interpreted cautiously.

Keywords Social media addiction, Occupational balance, Sleep quality, University students, Generation Z

Background

Generations are defined as cohorts of individuals born around the same years who share similar historical and social experiences, shaping comparable worldviews and behaviors [1]. Generation Z—here operationalized as individuals born in or after 2000—is widely described as the first generation to grow up in a fully digital environment [2]. As a result, Generation Z students engage intensively with smartphones and social media platforms and often prefer visual and instant forms of communication over traditional methods such as phone calls or email [3]. Although this technological fluency may offer

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functional advantages, it is also associated with emerging digital risks [4]. In particular, excessive technology engagement has been linked to behavioral addiction-like patterns, with social media addiction frequently discussed in this context. While it is not formally recognized as a disorder in DSM-5, problematic social media use has been described through addiction components such as salience, mood modification, tolerance, withdrawal, conflict, and relapse [5, 6].

Uncontrolled social media use can increasingly dominate daily routines and has been associated with musculoskeletal complaints, sleep disturbances, irritability, and reduced productivity [7, 8]. Sleep quality is a particularly critical outcome, given its central role in physical and mental health; nighttime screen exposure may delay sleep onset and reduce sleep duration and quality, contributing to fatigue, attentional difficulties, and cognitive impairments in young adults [9]. From an occupational therapy perspective, these patterns are also relevant to occupational balance, defined as maintaining a harmonious pattern of meaningful daily activities [10]. Disruptions in occupational balance are associated with lower life satisfaction, reduced performance, and poorer mental well-being [11]. For Generation Z, greater time spent in digital environments may constrain balance across social, academic, rest, and self-care domains, potentially undermining overall quality of life [12, 13]. Consistent with this view, recent findings have reported a positive correlation between sleep duration and occupational balance, and a negative correlation between smartphone usage time and occupational balance among young adults [14]. Therefore, this study investigates the relationship between social media addiction, sleep quality, and occupational balance among Generation Z university students, aiming to inform preventive strategies that support healthier digital habits and more balanced daily routines.

Materials and methods

This study was designed to examine the relationship between social media addiction, sleep quality, and occupational balance among Generation Z university students. This study was approved by the Hamidiye Scientific Research Ethics Committee of University of Health Sciences, Türkiye (Approval No: 23/7-7/1; Date: 07 April 2023), and was conducted in accordance with the principles of the Declaration of Helsinki. The sample consisted of 136 participants who were currently enrolled in a university program and were born in or after the year 2000. The minimum required sample size was calculated as 134 based on a 0.05 margin of error and 0.95 confidence level. The participants were recruited using a convenience sampling method. Data collection was facilitated through digital platforms such as university-based

social media and messaging groups to reach eligible and accessible participants efficiently.

Inclusion criteria for the study were as follows: volunteering to participate in the research, being born in or after 2000, and currently studying at a university. Exclusion criteria included having a diagnosed sleep disorder (e.g., insomnia, parasomnia, hypersomnia, or somnambulism); use of psychiatric medication; chronic musculoskeletal pain; and sleep problems caused by chronic conditions such as diabetes, cardiovascular disease, or hypertension.

Measures

Data were collected using a sociodemographic information form and three standardized measurement tools.

Sociodemographic form

Participants were asked to complete a sociodemographic questionnaire designed to collect background information relevant to the study variables. The form included the following items: participants' name and surname, gender, age (in years), the name of the university attended, estimated daily social media usage (in hours), the most frequently used social media platform, and the current year of study (academic grade level).

Social Media Addiction Scale (SMAS)

Social media addiction was assessed using the 41 item likert-type Social Media Addiction Scale developed for university students [15]. The scale consists of 41 items and is a 5-point likert-type measurement tool, where participants respond using frequency expressions ranging from "always," "often," "sometimes," "rarely," to "never." The SMAS demonstrates a four-factor structure (occupation, mood modification, relapse, and conflict). The internal consistency coefficient of the scale was reported as Cronbach's $\alpha = 0.967$. The total score ranges from 41 to 205, with higher scores indicating higher levels of social media addiction. Items 1–12 measure the "Occupation" dimension, which evaluates how much social media occupies the individual. Items 13–17 are related to the "Mood Modification" dimension, assessing the extent to which social media influences emotional states. Items 18–22 represent the "Relapse" dimension, evaluating the tendency to return to social media use. Items 23–41 focus on the "Conflict" dimension, measuring the degree to which social media use causes negative consequences in the individual's life. The scale includes four factors and uses a 5-point rating system ranging from "Never" to "Always." Higher scores indicate higher levels of addiction [15].

Pittsburgh Sleep Quality Index (PSQI)

Sleep quality was measured using the Pittsburgh Sleep Quality Index. The PSQI includes 19 self-rated items grouped into 7 components evaluating subjective sleep quality, latency, duration, efficiency, disturbances, medication use, and daytime dysfunction. Total scores range from 0 to 21, with higher scores indicating poorer sleep quality. Scores of 5 and above are considered indicative of the presence of a sleep disorder. In this study, the Turkish version of the PSQI showed acceptable reliability (Cronbach's $\alpha = 0.722$), aligning with the original Turkish adaptation (Cronbach's $\alpha = 0.80$) [16].

Occupational Balance Questionnaire (OBQ)

Occupational balance was assessed using the Turkish version of the Occupational Balance Questionnaire developed by Dür et al. [17]. This 11-item, single-factor scale measures perceived balance between different types of daily occupations. Each item is rated on a 4-point Likert scale, and the total score ranges from 0 to 33, with higher scores reflecting better occupational balance. The internal consistency in this study was strong (Cronbach's $\alpha = 0.828$), consistent with the Turkish adaptation study ($\alpha = 0.83$) [18].

Data analysis

All statistical analyses were conducted using IBM SPSS Statistics version 25.0. Descriptive statistics, including means, standard deviations, frequencies, and percentages, were used to summarize participant characteristics and scale scores. The normality of data distribution was evaluated using skewness and kurtosis values. Independent samples t-tests were applied to compare means between two groups, while one-way analysis of variance (ANOVA) was used for comparisons across more than two groups. Post hoc analyses were conducted using the Bonferroni correction method to identify specific group

differences when ANOVA results were significant. In addition, Pearson correlation analysis was performed to examine the relationships between social media addiction, sleep quality, and occupational balance scores. Pearson correlation analysis was performed to examine the relationships among social media addiction, sleep quality, and occupational balance scores. To reduce the risk of Type I error due to multiple correlation analyses, FDR-adjusted p-values were calculated using the Benjamini–Hochberg false discovery rate procedure. In addition, multiple linear regression analyses were conducted to examine the factors associated with sleep quality and occupational balance after adjusting for available potential confounders. In the first model, Pittsburgh Sleep Quality Index score was entered as the dependent variable, while age, gender, academic year, daily social media use duration, and total social media addiction score were entered as independent variables. In the second model, occupational balance total score was entered as the dependent variable, while age, gender, academic year, daily social media use duration, total social media addiction score, and Pittsburgh Sleep Quality Index score were entered as independent variables. Multicollinearity was assessed using variance inflation factor values. All statistical analyses were conducted using two-tailed tests, and a p-value of $p < .05$ was considered statistically significant.

Results

Participant socio-demographic characteristics

A total of 136 university students participated in the study (M age = 21.03 years, $SD = 1.17$). The sample predominantly comprised female students, with the majority enrolled in their third or fourth year of study. Instagram emerged as the most frequently reported primary social media platform, and the most common daily usage duration was within the range of 1–3 h. Comprehensive demographic characteristics and patterns of social media use are presented in Table 1.

Descriptive analysis of occupational balance, sleep quality and social media addiction

The mean occupational balance score of the participants was 18.19 ($SD = 2.72$). According to the Pittsburgh Sleep Quality Index (PSQI), where scores of 5 or greater indicate sleep disturbances, the mean score was 5.81 ($SD = 3.14$). Overall, 60.3% ($n = 82$) of the participants were classified as having poor sleep quality, whereas 39.7% ($n = 54$) were classified as having good sleep quality. Detailed descriptive statistics for all variables are presented in Table 2.

Table 1 Demographic characteristics of the participants and social media use

Variables	Category	n	%
Gender	Female	106	77.9
	Male	30	22.1
Year of Study	First Year	22	16.2
	Second Year	23	16.9
	Third Year	40	29.4
	Fourth Year	51	37.5
Most Frequently Used Platform	Instagram	100	73.5
	Twitter	15	11.0
	WhatsApp	10	7.4
	YouTube	11	8.1
Daily Usage Duration	1-3	69	50.7
	4-6	56	41.2
	More than 6 h	11	8.1

Table 2 Descriptive analysis of occupational balance, sleep quality, and social media addiction

Variables	N	Mean	SD
Age	136	21.03	1.17
Occupational Balance	136	18.19	2.72
Pittsburgh Sleep Quality Index	136	5.81	3.14
· Poor Sleep	82	7.77	2.45
· Good Sleep	54	2.83	1
Social Media Addiction	136	99.05	26.46
· Occupation	136	36.6	9.09
· Mood Modification	136	14.26	4.84
· Relapse	136	10.96	4.47
· Conflict	136	37.23	12.41

Analysis of differences in social media addiction scores by demographic variables

An independent samples *t*-test showed no significant gender differences in SMAS total scores. $t(40.57) = -0.83$, $p = .409$. A one-way analysis of variance (ANOVA) was performed to evaluate whether there were significant differences in the SMAS total scores across the years of study. The results indicated that the differences among years of study were not statistically significant. $F(3, 132) = 1.14$, $p = .337$.

Analysis of differences in social media addiction, occupational balance, and sleep quality scores by social media use variables

A significant difference was observed in social media addiction scores according to daily social media usage duration ($F = 7.57$, $p < .001$; Table 3). Bonferroni post-hoc tests showed that students who used social media for more than six hours per day had significantly higher.

scores than those using it for 1–3 h or 4–6 h. In contrast, no significant differences were observed across the groups for PSQI and occupational balance.

A significant difference was found between students with good and poor sleep quality in terms of social media addiction scores. Students with poor sleep quality had higher Social Media Addiction Scale scores than those with good sleep quality, with a medium effect size (good sleep: 91.15 ± 24.92 ; poor sleep: 104.26 ± 26.29 ; mean difference = -13.11 , 95% CI $[-22.04, -4.18]$; $t = -2.90$, $p = .004$; Cohen's $d = -0.51$; Table 4). Similarly, students with poor sleep quality had significantly higher Occupational Balance Questionnaire scores than those with good sleep quality, with a large effect size (good sleep: 16.93 ± 2.33 ; poor sleep: 19.02 ± 2.65 ; mean difference = -2.09 , 95%

Table 3 Comparison of Social Media Addiction, Sleep Quality (PSQI), and occupational balance across daily social media usage groups: one-way ANOVA

		Daily Social Media Usage	n	Mean	SD	F	p	Multiple Comparison
Social Media Addiction	1	1–3 h	69	93.45	26.39	7.57	< 0.001	3>1
	2	4–6 h	56	100.89	23.76			
	3	More Than 6 h	11	124.82	25.33			3>2
PSQI	1	1–3 h	69	5.8	3.25	0.19	0.82	
	2	4–6 h	56	5.93	2.92			
	3	More Than 6 h	11	5.27	3.66			
Occupational Balance	1	1–3 h	69	18.07	2.88	0.21	0.80	
	2	4–6 h	56	18.38	2.59			
	3	More Than 6 h	11	18	2.53			

Bold values indicate statistical significance at $p < .05$

The “more than 6 hours” group included a relatively small number of participants; therefore, comparisons involving this subgroup should be interpreted with caution

$p < 0.05$

Table 4 A comparison of the social media addiction scores and occupational balance questionnaire between good sleep and poor sleep

Variable	Good Sleep (n=54)	Poor Sleep (n=82)	Mean Difference 95% CI	t	p	Cohen's d
	Means $\pm$ SD	Means $\pm$ SD				
Social Media Addiction	91,15 $\pm$ 24,92	104,26 $\pm$ 26,29	-13.11 [-22.04, -4.18]	-2,9	0.004	-0.51
Occupational Balance	16,93 $\pm$ 2,33	19,02 $\pm$ 2,65	-2.09 [-2.97, -1.21]	-4,73	<0.001	-0.83

Bold values indicate statistical significance at $p < .05$

PSQI ≥ 5 Poor Sleep, PSQI < 5 Good Sleep

Table 5 Relationships among social media addiction, occupational balance, and sleep quality

Variables	Pittsburgh Sleep Quality Index Score	Occupational Balance Total
Occupation	0.219* (0.016)	0.279** (0.003)
Mood Modification	0.112 (0.212)	0.044 (0.611)
Relapse	0.136 (0.139)	0.289** (0.003)
Conflict	0.213* (0.018)	0.261** (0.004)
Social Media Addiction Score Total	0.219* (0.016)	0.275** (0.003)
Pittsburgh Sleep Quality Index Score	—	0.419** (0.003)

Bold values indicate statistical significance at $p < .05$

Values are presented as Pearson correlation coefficients with FDR-adjusted p -values in parentheses. FDR-adjusted p -values were calculated using the Benjamini–Hochberg false discovery rate procedure

*FDR-adjusted $p < .05$; **FDR-adjusted $p < .01$

CI [-2.97, -1.21]; $t = -4.73$, $p < .001$; Cohen's $d = -0.83$; Table 4).

Pearson correlation analyses were performed, and FDR-adjusted p -values were calculated using the Benjamini–Hochberg false discovery rate procedure. After FDR correction, Pittsburgh Sleep Quality Index score was weakly and positively correlated with the occupation subscale of social media addiction ($r = .219$, FDR-adjusted $p = .016$), the conflict subscale ($r = .213$, FDR-adjusted $p = .018$), and the total social media addiction score ($r = .219$, FDR-adjusted $p = .016$). No significant correlations were found between Pittsburgh Sleep Quality Index score and mood modification or relapse after FDR correction. Occupational balance total score was weakly and positively correlated with the occupation subscale ($r = .279$, FDR-adjusted $p = .003$), relapse subscale ($r = .289$, FDR-adjusted $p = .003$), conflict subscale

($r = .261$, FDR-adjusted $p = .004$), and total social media addiction score ($r = .275$, FDR-adjusted $p = .003$). Mood modification was not significantly correlated with occupational balance total score. In addition, Pittsburgh Sleep Quality Index score showed a moderate positive correlation with occupational balance total score ($r = .419$, FDR-adjusted $p = .003$). Overall, although several correlations remained statistically significant after FDR correction, the associations between social media addiction variables and sleep quality or occupational balance were generally weak in magnitude and should be interpreted cautiously. (Table 5)

Multiple linear regression analyses were performed to examine the factors associated with sleep quality and occupational balance. The results are presented in Table 6. In the first model, PSQI score was entered as the dependent variable, while age, gender, academic year, daily social media use duration, and total social media addiction score were included as independent variables. The overall model was not statistically significant, $F(5,130) = 1.911$, $p = .097$, and explained 6.8% of the variance in PSQI scores. Among the predictors, total social media addiction score was positively associated with PSQI score ($B = 0.030$, $\beta = 0.255$, $p = .007$), indicating that higher social media addiction scores were associated with higher PSQI scores after controlling for age, gender, academic year, and daily social media use duration. Age, gender, academic year, and daily social media use duration were not significant predictors of PSQI score.

In the second model, occupational balance total score was entered as the dependent variable, while age, gender, academic year, daily social media use duration, total social media addiction score, and PSQI score were included as independent variables. The overall model was statistically significant, $F(6,129) = 6.543$, $p < .001$, and explained 23.3% of the variance in occupational balance scores. Among the predictors, only PSQI score was

Table 6 Multiple Linear Regression Analyses Predicting Sleep Quality and Occupational Balance

Variable	PSQI Score B	SE	β	p	VIF	Occupational Balance B	SE	β	p	VIF
Age	0.068	0.309	0.026	0.825	1.925	0.420	0.244	0.184	0.087	1.925
Gender	0.340	0.677	0.045	0.616	1.122	0.454	0.535	0.069	0.397	1.124
Academic year	0.120	0.325	0.041	0.714	1.774	-0.147	0.257	-0.059	0.567	1.776
Daily social media use duration	-0.215	0.152	-0.132	0.159	1.205	0.025	0.121	0.018	0.836	1.223
Social media addiction total score	0.030	0.011	0.255	0.007	1.214	0.016	0.009	0.154	0.081	1.284
PSQI score	—	—	—	—	—	0.320	0.069	0.370	<0.001	1.074
R^2	0.068					0.233				
Adjusted R^2	0.033					0.198				
F	1.911					6.543				
p	0.097					<0.001				

Bold values indicate statistical significance at $p < .05$

PSQI score was included as a predictor only in the occupational balance model. P -values reported as 0.000 in SPSS were presented as $p < .001$. All VIF values were below 5, indicating no serious multicollinearity

B unstandardized regression coefficient, SE standard error, β standardized regression coefficient, VIF variance inflation factor, PSQI Pittsburgh Sleep Quality Index

significantly and positively associated with occupational balance total score ($B = 0.320$, $\beta = 0.370$, $p < .001$). Total social media addiction score was not a significant predictor of occupational balance after controlling for PSQI score and the other covariates. Age, gender, academic year, and daily social media use duration were also not significant predictors. All VIF values were below 5, indicating no serious multicollinearity.

Discussion

This study examined the relationships among social media addiction, sleep quality, and occupational balance in university students. The findings showed that students with higher social media addiction scores tended to report poorer sleep quality. Social media addiction scores also differed according to daily usage duration, with students using social media for more than six hours per day reporting the highest addiction scores. Instagram was the most frequently used social media platform among the participants. Although a weak positive association was found between social media addiction and occupational balance in the bivariate analyses, this association did not remain significant in the multivariable analysis. This finding suggests that the relationship between social media addiction and occupational balance may not be direct or strong, but rather should be considered together with sleep quality and individual characteristics. Overall, the results indicate that the relationships among social media use, sleep quality, and perceived occupational balance are multidimensional.

This study found that social media addiction scores differed significantly according to daily social media use duration. In particular, students who reported using social media for more than six hours per day had higher social media addiction scores than those who reported using it for 1–3 h or 4–6 h per day. This finding suggests that duration of use is an important factor associated with social media addiction. However, daily usage duration alone may not fully explain addictive patterns of social media use; factors such as purpose of use, type of online interaction, psychosocial stress, coping strategies, and emotion regulation should also be considered. In addition, the number of participants in the “more than six hours” group was smaller than in the other daily use groups. Therefore, while the ANOVA and post-hoc analyses provide meaningful preliminary evidence regarding differences in social media addiction according to daily usage duration, comparisons involving this subgroup should be interpreted with caution. Future studies with larger and more balanced subgroup distributions are recommended to further confirm these findings. In contrast, no significant differences in social media addiction scores were found according to gender or academic year. Although previous studies have suggested that women

may be more vulnerable to problematic social media use due to their tendency to engage more intensively in social interaction and self-expression via social media, [19, 20] this pattern was not observed in the present study. However, because the sample was predominantly female, the interpretation of gender-based comparisons is limited. Therefore, the absence of a significant gender difference should not be interpreted as evidence that gender has no role in social media addiction, but rather as a finding specific to the present sample. Overall, these findings indicate that daily usage duration and usage habits are important when evaluating social media addiction among university students, while results related to demographic variables should be interpreted in light of the sample characteristics.

In this study, Instagram was reported as the most frequently used social media platform among the participants. However, the present findings showed that social media addiction scores differed particularly according to daily usage duration, with students using social media for more than six hours per day reporting higher addiction scores. When interpreting the findings related to platform preference, it should be considered that the majority of participants used Instagram, whereas the number of students preferring other platforms was relatively limited. This distribution may have restricted the ability to compare addiction levels across different platforms. Although previous studies have suggested that visually oriented and socially interactive platforms such as Instagram may pose a risk for problematic use [6], the findings of the present study suggest that social media addiction may be more closely associated with daily usage duration and usage patterns than with platform type. Therefore, the results related to platform preference should be interpreted in light of the sample distribution.

The positive association identified between social media addiction and poorer sleep quality in this study is consistent with previous research showing that problematic social media use is associated with sleep disturbances. Abu-Snieneh et al. (2020) reported that individuals with higher levels of social media dependence experienced significantly poorer sleep quality and more pronounced daytime dysfunction. Similarly, Angelillo et al. [21] emphasized that frequent social media engagement was one of the main predictors of sleep disturbances. Additionally, Zhu et al. [19] demonstrated that social media addiction and social media fatigue significantly contributed to deteriorating sleep quality, while fear of missing out further amplified the impact of problematic use. In line with these findings, the present study showed that higher social media addiction scores were associated with higher PSQI scores, indicating poorer sleep quality. This association also remained significant in the regression analysis after controlling for age, gender,

academic year, and daily social media use duration. The relationship may be related to behavioral patterns such as prolonged evening screen exposure, delayed bedtime, and compulsive checking; however, these mechanisms were not directly assessed in the present study. Therefore, the findings should be interpreted as evidence of an association rather than a causal relationship. From a public health perspective, these findings highlight the importance of considering problematic social media use as a behavioral factor related to sleep health among university students.

In the bivariate analyses, social media addiction was weakly and positively associated with occupational balance; however, this association did not remain significant in the multivariable regression model after adjusting for sleep quality, age, gender, academic year, and daily social media use duration. This finding suggests that the relationship between social media addiction and occupational balance may be indirect or influenced by other individual and behavioral factors. From an occupational therapy perspective, occupational balance is not only related to the distribution and variety of daily activities, but also to individuals' subjective perceptions of having the right amount and variation of occupations, as well as satisfaction with daily occupational patterns [11, 14, 17]. Therefore, the initial positive association may reflect students' subjective appraisal of digitally mediated routines rather than an objectively healthy or balanced occupational pattern. Since the present study did not directly assess mechanisms such as immediate gratification, emotional regulation, perceived social connectedness, or coping strategies, this finding should not be interpreted as evidence that social media addiction improves occupational balance. Future longitudinal and qualitative studies are needed to clarify the direction and underlying mechanisms of these relationships.

This study has several limitations that should be considered when interpreting the findings. First, the cross-sectional design does not allow causal inferences to be made regarding the relationships among social media addiction, sleep quality, and occupational balance. Therefore, the findings should be interpreted as associations rather than evidence of directional or causal effects. Second, the use of convenience sampling may limit the representativeness of the sample and reduce the generalisability of the findings to the broader Generation Z university student population. In addition, the sample was female-dominant, which may have influenced the observed associations and limits the extent to which the findings can be generalised across gender groups. Third, although the overall sample size met the minimum required sample size based on the predefined margin of error and confidence level, some subgroup analyses included relatively small and unequal group sizes. In particular, the

subgroup of participants who reported using social media for more than six hours per day included only 11 individuals. This may have limited the statistical robustness of the ANOVA and post hoc comparisons and reduced the reliability of subgroup-based interpretations. Therefore, subgroup findings should be interpreted with caution. Fourth, the use of self-report questionnaires may have introduced recall or response bias. In addition, sleep quality and screen time were not assessed using objective measures, which may limit the precision of the reported behavioural data. Finally, unmeasured factors such as mental health status, academic workload, socioeconomic status, physical activity, and chronotype may have influenced the observed associations.

Despite these limitations, the study contributes to the literature by examining social media addiction, sleep quality, and occupational balance together from an occupational therapy perspective. Future studies should use longitudinal or experimental designs, larger and more gender-balanced samples, objective sleep and screen-time measurements, and more comprehensive assessment of potential confounders to strengthen the evidence base.

Conclusion

Social media addiction was associated with sleep quality among Generation Z university students, and this association remained significant after adjustment for sociodemographic and social media use-related variables. Although social media addiction showed a weak positive association with occupational balance in bivariate analyses, this relationship did not remain significant in the multivariable model after sleep quality and other covariates were considered. Therefore, the relationship between social media addiction and occupational balance should be interpreted cautiously and may be influenced by sleep quality and other unmeasured individual or contextual factors.

From a public mental health and occupational therapy perspective, these findings suggest that problematic social media use should be considered as a behavioral factor related to sleep health and daily routine regulation among university students. Preventive strategies aimed at promoting digital well-being, healthy sleep habits, and balanced engagement in daily occupations may support students' overall well-being.

Abbreviations

SMAS Social Media Addiction Scale
PSQI Pittsburgh Sleep Quality Index
OBQ Occupational Balance Questionnaire

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Authors' contributions

HYD: Conceptualization, Methodology, Data collection, Formal analysis, Writing – original draft. ZB: Supervision, Writing – review & editing. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study was approved by the Hamidiye Scientific Research Ethics Committee of University of Health Sciences, Türkiye (Approval No: 23/7–7/1; Date: 07 April 2023). All participants provided informed consent prior to participation in the study. The research was conducted in accordance with the principles of the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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